

Database Introduction (DBI202)

Objectives :

- ✓ Understand the database concepts and database management system software
- ✓ Understand the relation model of data and Algebraic Query Language
- ✓ Understand data normalization and apply normalization techniques in database design
- ✓ Be able to model an application's data requirements using conceptual modeling tools like ER diagrams and design database schemas based on the conceptual model.
- ✓ Be proficient in structure query language including Data Definition Language(DDL) and Data Manipulation Language(DML)
- ✓ Understand PL/SQL concepts and manipulate with View, Cursors, Stored Procedures, Functions, Database Triggers
- ✓ Apply the Index in database design and query optimization
- ✓ Apply AI Tools and Computational Thinking (CT) tools to identify, analyze, model, and demonstrate problem-solving

1. Smart Parking Management System

Context

Urbanization and rapid growth in vehicle ownership have created significant challenges in managing parking infrastructure efficiently. Traditional parking systems often rely on manual monitoring and static allocation of parking spaces, resulting in congestion, underutilization of resources, and poor user experience.

A Smart Parking Management System aims to address these issues by integrating real-time data collection, automation, and centralized data management. The system leverages IoT-based sensors to continuously monitor parking space availability and update status dynamically. It supports end-to-end parking operations, including reservation, entry/exit tracking, and automated billing. By maintaining consistent and accurate records of vehicles, parking sessions, and payments, the system enables efficient resource allocation and transparency in operations.

Furthermore, the system generates analytical reports to support decision-making, such as identifying peak usage periods and optimizing pricing strategies. This project provides a realistic scenario for designing a database that handles transactional data, real-time updates, and operational analytics in a smart city context.

Proposal Solution

The proposed solution is a centralized database-driven system integrated with IoT sensors to enable real-time monitoring of parking availability. It supports reservation, automated check-in/check-out, and billing processes. The system ensures data consistency through transactional control while providing analytical dashboards to optimize parking utilization and pricing strategies.

Actors & Features

- **Driver**

- Search for available parking spots.
- Reserve parking spaces in advance.
- Check in and check out of parking areas.
- Make payments for parking sessions.

- **Parking Staff**

- Monitor vehicle entry and exit.
- Assist drivers when needed.

- **System Sensor**

- Detect parking space availability.
- Update real-time status of parking spots.

- **Administrator**

- Manage parking lots and parking spaces.
- Configure pricing rules (hourly/daily rates).
- Monitor system performance and reports.

Business Rules

- A parking spot can accommodate **only one vehicle at a time**.
- A vehicle may have multiple parking sessions over time.
- Each parking session is linked to one vehicle and one parking spot.
- Parking fees are calculated based on the duration of the parking session.
- A parking spot cannot be reserved or occupied if it is already in use.
- Each parking session must have a corresponding payment record.

Reports / Statistics

- Parking occupancy report (available vs occupied spots).
- Vehicle parking history.
- Revenue report (daily, weekly, monthly).
- Parking lot utilization rate.

2. Food Delivery System

Context

The rapid expansion of digital platforms and mobile applications has transformed the food service industry, making online food delivery an essential service in modern urban life. Food delivery systems must coordinate multiple stakeholders, including customers, restaurants, and delivery drivers, while ensuring timely and accurate order fulfillment.

These systems handle a high volume of concurrent transactions, requiring robust database design to maintain consistency, scalability, and reliability. Key challenges include managing dynamic menus, real-time order tracking, delivery logistics, and secure payment processing.

Additionally, the system must support efficient communication between participants and provide transparency in order status updates. Data collected from user interactions and transactions can be further utilized for analytics, such as identifying popular food items, optimizing delivery routes, and improving service quality.

This project offers students an opportunity to design a comprehensive database system that supports complex workflows, real-time operations, and data-driven decision-making in a highly competitive digital marketplace.

Proposal Solution

The proposed solution is a scalable, database-centric platform that coordinates customers, restaurants, and delivery drivers through real-time data synchronization. It supports order lifecycle management, dynamic menu updates, and secure payment processing. The system incorporates efficient transaction handling and data analytics to improve delivery performance, customer experience, and operational decision-making.

Actors & Features

• Customer

- Browse restaurants and menus.
- Place food orders.
- Track order status in real time.
- Make online payments.

• Restaurant

- Manage menu items and pricing.

- Receive and process orders.
- Update order preparation status.

• **Delivery Driver**

- Accept delivery requests.
- Pick up orders from restaurants.
- Deliver orders to customers.
- Update delivery status.

• **Administrator**

- Manage users and system data.
- Monitor transactions and performance.
- Handle complaints and support requests.

Business Rules

- An order is placed by **one customer** and may contain multiple food items.
- A restaurant can receive multiple orders.
- A delivery driver can handle multiple deliveries, but only one active delivery at a time.
- Each order must be assigned to one restaurant and one delivery driver.
- Payments are linked one-to-one with orders.
- An order cannot be delivered unless it has been confirmed and prepared by the restaurant.

Reports / Statistics

- Order history report by customer.
- Revenue report by restaurant.
- Delivery performance report (time, success rate).
- Most popular food items report.

3. Cosplay Costume Rental Management System

Context

The rapid growth of entertainment culture, including anime, comics, gaming, and cosplay events, has led to an increasing demand for costume rental services. Individuals often prefer renting cosplay costumes instead of purchasing them due to high costs and limited usage frequency. However, many existing rental services operate manually or with fragmented systems, leading to inefficiencies in inventory tracking, booking management, and customer service.

A Cosplay Costume Rental Management System aims to digitalize and streamline the entire rental process, including costume catalog management, reservations, rental tracking, and payment processing. The system must handle diverse data types such as costume attributes (size, character, theme), user preferences, rental history, and media (images).

Additionally, it supports real-time availability updates and integrates analytics to optimize inventory usage. This project provides a practical scenario for designing a database system that manages transactional workflows, inventory control, and user interactions in a niche but growing digital marketplace.

Proposal Solution

The proposed solution is a web-based, database-driven system that manages costume inventory, rental transactions, and customer interactions in a centralized platform. It ensures data consistency through relational modeling and transaction management while supporting real-time availability tracking.

Actors & Features

• Customer

- Browse cosplay costumes by category, character, size, or theme.
- View costume details, images, and availability status.
- Make rental reservations for specific dates.
- Manage rental orders (extend, cancel, or return).
- Make online payments and view payment history.

• Shop Staff

- Manage costume inventory (add, update, remove costumes).
- Confirm and process rental orders.
- Handle costume check-out and return processes.
- Inspect costume condition before and after rental.
- Update costume status (available, rented, maintenance).

• Administrator

- Manage users (customers, staff).
- Configure pricing policies (daily rate, late fees, deposit rules).
- Monitor system transactions and performance.
- Generate reports and analytics.

Business Rules

- Each costume has a unique ID and can exist in multiple sizes and variants.
- A costume can only be rented to one customer during a specific time period.
- A customer may have multiple rental orders but cannot exceed a predefined limit (e.g., 3 active rentals).
- Each rental order must include at least one costume and specify a rental duration.
- Rental fees are calculated based on duration, costume category, and pricing rules.
- Late returns incur additional fees calculated per overdue day.
- Each rental transaction must have a corresponding payment record.
- Costumes under maintenance cannot be reserved or rented.

Reports / Statistics

- Costume utilization report (rented vs available items).
- Rental history by customer.
- Revenue report (daily, weekly, monthly).
- Most popular cosplay characters/themes.
- Late return and penalty report.
- Inventory condition and maintenance report.

4. Household Cleaning Robot Sales & Maintenance Management System

Context

The rapid adoption of smart home technologies has significantly increased the demand for household cleaning robots such as robotic vacuum cleaners and floor-mopping devices. Consumers not only require efficient purchasing channels but also reliable after-sales services, including maintenance, repair, and warranty management. However, many small and medium-sized businesses still manage these operations using disconnected systems or manual processes, leading to inefficiencies in inventory control, service scheduling, and customer support.

A centralized system is required to manage the entire lifecycle of cleaning robots—from sales transactions and customer records to maintenance history and service operations.

The system must handle diverse data types, including product specifications, warranty information, service logs, and customer interactions. Additionally, integrating IoT data from smart robots (e.g., usage statistics, error logs) can enhance predictive maintenance and service quality.

This project provides a realistic case for designing a database system that supports transactional workflows, asset tracking, and data-driven service optimization in the smart home ecosystem.

Proposal Solution

The proposed solution is a centralized, database-driven system that integrates product sales, customer management, and maintenance services for household cleaning robots. It ensures data consistency through relational database design and transaction management while supporting real-time updates and service tracking. The system can be extended with IoT data integration and AI-based predictive maintenance to improve operational efficiency and customer satisfaction.

Actors & Features

• Customer

- Browse available cleaning robots by brand, features, and price.
- View product specifications, warranty details, and user reviews.
- Purchase robots and track order status.
- Register purchased devices for warranty services.
- Request maintenance or repair services.
- Track service status and maintenance history.

• Sales Staff

- Manage product catalog (add/update/remove robot models).
- Process sales orders and invoices.
- Assist customers in selecting suitable products.
- Update product availability and pricing.

• Technician (Maintenance Staff)

- Receive and manage service requests.
- Schedule maintenance or repair appointments.
- Record service details (issues, actions taken, parts replaced).
- Update device status (under maintenance, repaired, returned).

• System (IoT/AI Extension)

- Collect usage data and error logs from connected robots.
- Detect anomalies and suggest predictive maintenance.
- Recommend maintenance schedules based on usage patterns.

• Administrator

- Manage users (customers, staff, technicians).
- Configure pricing, warranty policies, and service fees.
- Monitor system performance and transactions.
- Generate analytical reports and insights.

Business Rules

- Each robot model has a unique product ID and may have multiple units in inventory.
- Each purchased robot is associated with exactly one customer and includes a warranty period.

- A customer may own multiple robots.
- Each service request must be linked to one specific robot.
- A robot can have multiple maintenance records over time.
- Service requests must be assigned to one technician.
- Maintenance cannot be performed if the robot is not registered in the system.
- Warranty coverage determines whether service fees are applied.
- Each sales transaction and service operation must have a corresponding payment record (if applicable).

Reports / Statistics

- Sales report by product and time period.
- Inventory status report (available, sold, under maintenance).
- Maintenance history report by customer or device.
- Technician performance report (tasks completed, time taken).
- Warranty claim and service cost report.
- Predictive maintenance and failure trend analysis.

5. Career Guidance Kiosk System for High School Students at FPT University Open Day (HCM Campus)

Context

University open day events play a crucial role in helping high school students explore academic programs and make informed career decisions. At large-scale events such as those organized by FPT University (Ho Chi Minh City campus), thousands of students interact with booths, counselors, and informational materials within a limited time.

Traditional consultation methods often rely on face-to-face interactions, which can lead to long waiting times, inconsistent information delivery, and limited personalization. A Career Guidance Kiosk System provides an interactive, self-service platform where students can input their interests, academic strengths, and career aspirations to receive personalized recommendations. The system integrates structured data (programs, admission criteria) and semi-structured data (student preferences, survey responses). It can also incorporate AI-based recommendation models to suggest suitable majors and career paths.

This project offers a realistic scenario for designing a database system that supports high-volume interactions, real-time querying, and data-driven decision support in an educational environment.

Proposal Solution

The proposed solution is an interactive kiosk-based system supported by a centralized database that manages student inputs, academic program data, and recommendation results. It ensures efficient data processing and real-time response for high user volumes. The system can be extended with AI-based recommendation algorithms and analytics dashboards to enhance personalized guidance and support strategic decision-making for university recruitment activities.

Actors & Features

• Student (High School Visitor)

- Interact with kiosk to input personal interests, skills, and academic preferences.
- Browse university programs, majors, and career pathways.
- Receive personalized recommendations for suitable majors.
- View detailed program information (curriculum, career outcomes, admission requirements).
- Save or export consultation results (QR code, email, or print).

• University Counselor

- Access student interaction data from kiosks.
- Review student profiles and recommendation results.
- Provide follow-up consultation based on system suggestions.
- Record feedback and consultation notes.

• System (AI/Data Processing Module)

- Analyze student input data and match with suitable programs.
- Generate recommendation scores based on predefined rules or AI models.
- Log interaction data for analytics and improvement.

• Administrator

- Manage academic program data (majors, descriptions, admission criteria).
- Configure recommendation rules and questionnaires.
- Monitor kiosk usage and system performance.
- Generate reports and analytics for event evaluation.

Business Rules

- Each student interaction session is uniquely identified and time-stamped.
- A student may complete multiple consultation sessions, but each session is independent.
- Each recommendation result must be based on the student's input responses.

- Academic programs must have clearly defined attributes (major, requirements, career paths).
- Recommendation scores must follow predefined criteria or algorithmic logic.
- Student personal data must be handled securely and used only for consultation purposes.
- Kiosk sessions automatically expire after a period of inactivity.

Reports / Statistics

- Number of student interactions per kiosk.
- Most recommended majors/programs.
- Student interest distribution (by field or career area).
- Peak usage time during the event.
- Conversion rate (students requesting further consultation).
- Feedback and satisfaction report.

6. Capsule Hotel Rental Management System at Transportation Hubs

Context

With the rapid development of modern transportation systems such as airports, railway stations, and metro hubs, there is an increasing demand for short-term, flexible accommodation solutions for travelers.

Capsule (pod) hotels have emerged as an efficient alternative, offering compact, affordable sleeping spaces for passengers during transit, delays, or overnight stays. However, managing capsule rentals in high-traffic environments presents challenges, including real-time availability tracking, short-duration bookings, dynamic pricing, and rapid customer turnover.

Traditional management approaches are often inadequate for handling high concurrency and time-sensitive operations. A Capsule Hotel Rental Management System is designed to streamline booking, check-in/check-out, payment processing, and operational monitoring. The system must handle transactional data (bookings, payments), time-based availability, and potentially IoT data from smart capsules (e.g., occupancy sensors, energy usage).

This project provides a realistic case for designing a database system that supports time-based reservations, high-frequency transactions, and smart facility management in a smart transportation ecosystem.

Proposal Solution

The proposed solution is a centralized, database-driven system integrated with kiosk interfaces and IoT sensors to manage capsule rentals efficiently. It supports real-time booking, automated check-in/check-out, and dynamic pricing. The system ensures data consistency and scalability while enabling analytics and predictive insights to optimize utilization and operational performance.

Actors & Features

• Customer (Traveler)

- Search available capsules by location, time slot, and price.
- Book capsules for short-term stays (hourly or overnight).
- Check in and check out using QR code or kiosk system.
- Make online or on-site payments.
- View booking history and receive notifications.

• Front Desk Staff

- Assist customers with booking and check-in/check-out.
- Monitor capsule occupancy and availability.
- Handle customer issues or special requests.

• Maintenance Staff

- Update capsule status (available, occupied, cleaning, maintenance).
- Record cleaning schedules and maintenance logs.
- Report technical issues.

• System (IoT Extension)

- Detect occupancy using sensors.
- Monitor environmental conditions (temperature, usage time).
- Automatically update capsule status in real time.

• Administrator

- Manage capsule inventory and locations.
- Configure pricing policies (hourly, peak-time rates).
- Monitor system usage and financial performance.
- Generate operational and analytical reports.

Business Rules

- Each capsule has a unique ID and belongs to a specific location (station/zone).
- A capsule can only be occupied by one customer at a time.
- Bookings are time-based and cannot overlap for the same capsule.
- Customers must check in within a defined time window after booking; otherwise, the reservation is canceled.
- Each booking must have a corresponding payment record.
- Capsules marked as “maintenance” or “cleaning” cannot be booked.
- Pricing is calculated based on duration and may vary by time (peak/off-peak).
- A customer may have multiple bookings but cannot have overlapping active bookings.

Reports / Statistics

- Capsule occupancy rate (per hour/day).
- Booking and revenue reports (by time period and location).
- Customer usage patterns (duration, frequency).
- Maintenance and cleaning performance report.
- Peak usage time analysis.
- No-show and cancellation rate report.

7. Smart Tourism & Travel Companion Application

Context

The rapid growth of digital technologies and mobile platforms has significantly transformed the tourism industry, shifting from traditional travel planning to highly personalized and data-driven experiences. Modern travelers expect integrated services that support trip planning, booking, navigation, and real-time recommendations within a single application. However, existing tourism systems are often fragmented, requiring users to switch between multiple platforms for accommodation, transportation, and activity planning.

A Smart Tourism & Travel Companion Application aims to provide a unified platform that manages diverse datasets, including user preferences, itineraries, bookings, geolocation data, and user-generated content such as reviews and ratings. The system must handle both structured data (users, bookings, services) and semi-structured data (travel logs, recommendations).

Additionally, integrating AI-based recommendation engines and real-time data sources (e.g., weather, traffic) enhances user experience. This project provides a realistic scenario for designing a scalable database system that supports personalization, real-time updates, and complex interactions in a dynamic digital tourism ecosystem.

Proposal Solution

The proposed solution is a centralized, database-driven application that integrates trip planning, booking, and recommendation services into a unified platform. It ensures data consistency and supports real-time updates while handling diverse data types. The system can be enhanced with AI-based personalization and external data integration to deliver intelligent, context-aware travel experiences and optimize service utilization.

Actors & Features

• Tourist (User)

- Create and manage personal profiles and travel preferences.
- Search destinations, attractions, accommodations, and services.
- Plan trips by creating itineraries (multi-day schedules).
- Book services (hotels, transportation, tours).
- Receive real-time recommendations based on location and preferences.
- View maps, navigation routes, and nearby attractions.
- Submit reviews and ratings after trips.

• Service Provider (Hotel, Tour Operator, Transport Provider)

- Manage service listings (description, pricing, availability).
- Update real-time availability and pricing.
- Receive and process booking requests.
- Manage customer feedback and ratings.

• System (AI/Data Processing Module)

- Analyze user behavior and preferences to generate personalized recommendations.
- Integrate external data sources (e.g., weather, traffic conditions).
- Suggest optimized travel itineraries.
- Log user interactions for analytics and system improvement.

• Administrator

- Manage users and service providers.
- Monitor system performance and transactions.
- Configure recommendation rules and policies.
- Generate reports and analytics for business insights.

Business Rules

- Each user has a unique account and may create multiple trips.
- A trip itinerary consists of multiple activities scheduled over time.
- Each booking must be associated with one user and one service provider.

- Service providers can manage multiple services, each with its own availability and pricing.
- Bookings must not exceed available capacity (e.g., hotel rooms, tour slots).
- Reviews can only be submitted after a completed booking.
- Pricing may vary dynamically based on demand or season.
- Each booking must have a corresponding payment record.

Reports / Statistics

- Booking and revenue reports by service type and time period.
- Most popular destinations and attractions.
- User travel behavior and preference analysis.
- Service provider performance (ratings, bookings).
- Peak travel periods and demand trends.
- Recommendation effectiveness report.

8. Online Learning Management System (LMS)

Context

The rapid advancement of digital technologies and the increasing demand for flexible education have led to the widespread adoption of online learning platforms. Learning Management Systems (LMS) play a critical role in delivering educational content, managing courses, tracking student progress, and facilitating interaction between instructors and learners. However, many existing systems face challenges such as scalability, inconsistent data management, limited personalization, and inefficient assessment workflows.

An LMS must handle diverse data types, including structured data (users, courses, enrollments, grades), unstructured data (learning materials, submissions), and semi-structured interaction logs. Additionally, modern LMS platforms increasingly integrate AI-driven features such as personalized learning recommendations, automated grading, and performance analytics.

This project provides a realistic scenario for designing a robust database system that supports multi-role interactions, high concurrency, content management, and data-driven decision-making in an educational environment.

Proposal Solution

The proposed solution is a centralized, database-driven LMS that manages users, courses, learning content, and assessment workflows in a unified platform. It ensures data consistency and scalability through relational database design and transaction management. The system can be extended with AI-based recommendation and analytics modules to enhance personalized learning and improve educational outcomes.

Actors & Features

• Student

- Register and manage personal profile.
- Browse and enroll in available courses.
- Access learning materials (documents, videos, links).
- Submit assignments and take quizzes/exams.
- View grades, feedback, and progress reports.
- Participate in discussions and forums.

• Instructor

- Create and manage courses (content, schedule, syllabus).
- Upload learning materials and resources.
- Design assignments, quizzes, and exams.
- Grade student submissions and provide feedback.
- Monitor student performance and engagement.

• System (AI/Data Processing Module)

- Recommend courses or learning paths based on student performance and interests.
- Support automated grading for objective assessments (e.g., quizzes).
- Analyze learning behavior and engagement patterns.
- Log user interactions for analytics and system improvement.

• Administrator

- Manage users (students, instructors).
- Manage course catalog and system configurations.
- Monitor system usage, performance, and security.
- Generate reports and analytics.

Business Rules

- Each user must have a unique account and role (student, instructor, admin).
- A student can enroll in multiple courses, and a course can have multiple students (many-to-many relationship).
- Each course must be created and managed by one instructor.
- Assignments and assessments must have defined deadlines.
- Submissions after deadlines may be marked as late or rejected based on policy.
- Each submission must be associated with one student and one assignment.
- Grades must be recorded for each evaluated submission.
- Students can only access courses they are enrolled in.
- Each course must contain at least one learning module or material.

Reports / Statistics

- Student performance report (grades, progress).
- Course enrollment and completion rates.
- Instructor activity and course effectiveness.
- Assignment submission statistics (on-time vs late).
- System usage analytics (active users, session duration).
- AI recommendation effectiveness report.

9. Healthcare Appointment & Patient Record System

Context

The healthcare sector increasingly relies on digital systems to manage patient information, clinical workflows, and service delivery efficiently. Traditional paper-based or fragmented electronic systems often lead to data inconsistency, duplicated records, scheduling conflicts, and limited accessibility of patient information.

A Healthcare Appointment & Patient Record System aims to provide a centralized platform for managing patient data, appointment scheduling, medical histories, and clinical interactions. The system must handle sensitive and regulated data, requiring strong data integrity, privacy, and access control mechanisms. It also needs to support concurrent access by multiple stakeholders such as patients, doctors, and administrative staff. In addition to structured data (patient records, appointments, prescriptions), the system may store semi-structured or unstructured data such as medical notes or diagnostic reports. Integration with AI technologies can further support clinical decision-making, such as symptom analysis or risk prediction.

This project provides a realistic and complex scenario for designing a secure, scalable, and transaction-oriented database system in a healthcare environment.

Proposal Solution

The proposed solution is a secure, centralized, database-driven system that manages patient records, appointments, and clinical workflows. It ensures data integrity, privacy, and efficient transaction processing while supporting concurrent access by multiple users. The system can be extended with AI-based decision support and analytics to enhance healthcare service quality and operational efficiency.

Actors & Features

• Patient

- Register and manage personal profile.
- Book, reschedule, or cancel appointments.
- View appointment history and medical records.
- Receive notifications and reminders for appointments.
- Access prescriptions and treatment summaries.

• Doctor

- View daily appointment schedule.
- Access patient medical history and previous records.
- Record diagnoses, prescriptions, and treatment notes.
- Update patient records after consultations.
- Request additional tests or follow-up appointments.

• Receptionist / Medical Staff

- Manage appointment scheduling and patient check-in.
- Register new patients and update information.
- Assist in coordinating patient flow.
- Handle billing and administrative tasks.

• System (AI/Data Processing Module)

- Analyze patient symptoms and suggest possible conditions (decision support).
- Provide appointment recommendations based on doctor availability.
- Detect anomalies or high-risk cases from historical data.
- Log system interactions for analytics and improvement.

• Administrator

- Manage users (patients, doctors, staff).
- Configure system policies (working hours, appointment duration).
- Monitor system performance and security.
- Generate reports and analytics.

Business Rules

- Each patient must have a unique medical record ID.
- A patient may have multiple appointments over time.
- Each appointment must be associated with one patient and one doctor.
- Appointments must not overlap for the same doctor.
- Medical records must be updated only by authorized medical staff.
- Each consultation must result in a recorded diagnosis or note.
- Prescriptions must be linked to a specific consultation.
- Sensitive patient data must be protected and access-controlled.
- Billing records must be associated with completed appointments.

Reports / Statistics

- Appointment statistics (daily, weekly, monthly).
- Patient visit history and frequency.
- Doctor workload and utilization report.
- Treatment and diagnosis trends.
- Revenue and billing reports.
- Patient no-show and cancellation rate.

10. Mini Research Based Learning (optional)

Research Question: How can organizations design data lakes for drone and AV operational data?

Sau hoạt động này, sinh viên có thể:

- Hiểu sự khác biệt giữa:
 - Database vs Data Warehouse vs Data Lake
- Nhận diện các loại dữ liệu thực tế (real-world data)
- Thiết kế basic data architecture
- Phát triển kỹ năng:
 - Research
 - Critical thinking
 - Teamwork

GuideLine:

Bước 1: Problem Decomposition (Phân rã vấn đề)

Sinh viên cần hiểu các khái niệm cốt lõi:

- **Data Lake là gì?** (so với Database / Data Warehouse)
- **Drone & AV operational data gồm những gì?**
 - Sensor data (camera, LiDAR, GPS)
 - Telemetry data (speed, battery, position)
- **Big Data characteristics (5Vs):** Volume, Velocity, Variety, Veracity, Value

- **Data pipeline / Data architecture**

Bước 2: Sub-Problems (Câu hỏi con)

- **What types of data are generated by drones and AVs?**
 - **What challenges exist in storing this data?**
 - Real-time vs batch
 - Large size (video, sensor streams)
 - **What is a data lake and how is it different from traditional databases?**
 - **What architecture is suitable for a data lake?**
 - Storage layer
 - Processing layer
 - **How to ingest data into the data lake?**
 - **How to ensure data quality and governance?**
- How can the data be used (analytics / AI)?**

Bước 3: Research Tasks (Nhiệm vụ cho sinh viên)

- Chia thành các nhóm nhỏ (mỗi nhóm 2–3 sub-problems):
- **Task 1: Explore Data Types**
 - Tìm hiểu dữ liệu từ drone/AV
 - Ví dụ thực tế (video, GPS logs...)
- **Task 2: Compare Storage Approaches**
So sánh:
 - Traditional Database
 - Data Warehouse
 - Data Lake
- **Task 3: Investigate Data Lake Architecture**
Tìm hiểu các thành phần:
 - Data ingestion
 - Storage (cloud, distributed systems)
 - Processing (batch vs streaming)
- **Task 4: Tools & Technologies**
Tìm hiểu các công nghệ phổ biến:
 - Hadoop / Spark
 - Cloud storage (AWS S3, Azure Data Lake)
- **Task 5: Design Proposal**
Mỗi nhóm đề xuất:
 - 1 kiến trúc data lake đơn giản cho drone/AV

Research Process (Quy trình thực hiện)

Step 1: Understand the Problem

- Đọc câu hỏi
- Xác định keywords: *data lake, drone data, AV data*

Step 2: Literature Search

Tìm trên:

- Google Scholar
- Tech blogs
- Documentation (AWS, Microsoft)

Step 3: Information Filtering

Chọn nguồn:

- Có ví dụ thực tế
- Có sơ đồ kiến trúc

Step 4: Analysis & Synthesis

- So sánh các giải pháp
- Tóm tắt thành sơ đồ đơn giản

Step 5: Create Solution

Vẽ:

- Data flow diagram
- Data lake architecture

Step 6: Presentation

Trình bày:

- Problem → Solution → Justification

Deliverables (Sản phẩm nộp)

Mỗi nhóm nộp theo yêu cầu của milestones

Tuần	Mục tiêu chính	Sinh viên	Deliverable
Tuần 1	Sinh viên hiểu bài toán và bối cảnh	Đọc đề bài Xác định keywords Brainstorm trong nhóm	1 trang: Problem understanding List keywords (≥ 10 keywords)
Tuần 2	Nắm kiến thức nền	Tìm hiểu: - Data types (video, sensor, GPS...) - Database vs Data Warehouse vs Data Lake Học cách search tài liệu (Google Scholar, blog kỹ thuật)	Summary (1–2 trang): Các loại dữ liệu So sánh các hệ thống lưu trữ
Tuần 3	Biết chia nhỏ bài toán	Phân rã thành sub-problems: Storage challenges Data ingestion Data processing	Danh sách: ≥ 5 sub-problems Mỗi sub-problem có mô tả ngắn

		Thảo luận nhóm	
Tuần 4	Hiểu cách hệ thống hoạt động	Tìm hiểu: Data lake architecture Tools (Hadoop, Spark, Cloud storage) Phân tích: Batch vs Streaming	1 sơ đồ: Data lake architecture (basic) 1 trang giải thích
Tuần 5	Thiết kế giải pháp	Mỗi nhóm:Thiết kế data lake cho drone/AV Xác định:Data flow , Storage, Processing	Diagram hoàn chỉnh:Data pipelineMô tả (1–2 trang)
Tuần 6	Cải thiện và phản biện	Peer review giữa các nhóm Giảng viên feedback:Tính hợp lý,Tính khả thi	Version cải tiến: Diagram Explanation
Tuần 7	Tổng hợp và trình bày	Final report : Problem ,Research Solution , presentation , Q&A	Final report : Problem ,Research Solution , presentation , Q&A